注意电流方向！注意电压方向！注意电流方向！注意电压方向！注意方向！方向！方向

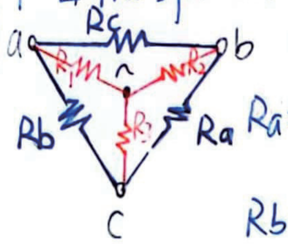
KCL：流入节点电流＝流出节点电流 注意电势差方向！ supernode：node间有电压源

KVL：环形电压升降为0 mesh：假设环形电流 → 实际每边电流 → KVL supermesh：mesh间有电流源

中间有不好处理的元件时假设其电流/电压，并添加元件特性方程.

Y-Δ Transformation:

如果题目给了Y：两个方向都有可能



Y→Δ:

$$R_a=\frac{R_1R_2+R_2R_3+R_1R_3}{R_1}$$

$$R_b=\frac{R_1R_2+R_2R_3+R_1R_3}{R_2}$$

$$R_c=\frac{R_1R_2+R_2R_3+R_1R_3}{R_3}$$

$$R_1R_2+R_1R_3+R_2R_3=\frac{R_aR_bR_c}{R_a+R_b+R_c}$$

Δ→Y:

$$R_1=\frac{R_bR_c}{R_a+R_b+R_c}$$

$$R_2=\frac{R_aR_c}{R_a+R_b+R_c}$$

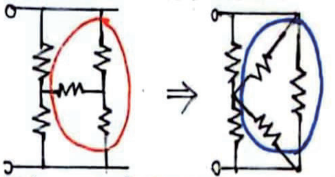
$$R_3=\frac{R_aR_b}{R_a+R_b+R_c}$$

$$R_{ab}(Y)=R_{ab}(\Delta)=R_1+R_2=\frac{R_c(R_a+R_b)}{R_a+R_b+R_c}$$

$$R_{ac}(Y)=R_{ac}(\Delta)=R_1+R_3=\frac{R_b(R_a+R_c)}{R_a+R_b+R_c}$$

$$R_{bc}(Y)=R_{bc}(\Delta)=R_2+R_3=\frac{R_a(R_b+R_c)}{R_a+R_b+R_c}$$

eg:



化简"横在中间的"电阻

★找出所有来源/去路 不想漏 找全两点间的所有可能通路

(Superposition principle) equivalent

叠加定理：一次只开一个 电压源关闭—短路 电流源关闭—开路 受控源都留下

Thevenin电路：电压源与电阻串联 ★注意电压方向(正负)

★非线性元件不能叠加 功率不能叠加

求解：$V_{TH}=V_{oc}$ (开路电压) R_{TH}：(法一) 无受控源时关闭所有独立源，两端之等效电阻 (法二)(有受控源可用) 计算短路电流 $R_{TH}=\frac{V_{oc}}{I_{sc}}$ (法三)(有受控源可用) 关所有独立源，加外加电压 V_{ex} 计算 I_{ex} $R_{TH}=\frac{V_{ex}}{I_{ex}}$ (法四) 找V-I curve上任意两点

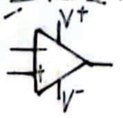
Norton电路：电流源与电阻并联 ★注意电流方向！一定检验！

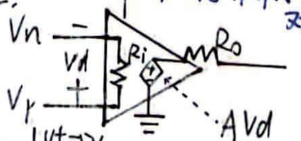
$I_N=\frac{V_{TH}}{R_{TH}}$ $R_N=R_{TH}$

电压源并联电阻/电流源串联电阻 对外电路无影响

最大功率传输：外电路电阻 $R_L=R_{TH}$

运算放大器 Op-Amp.

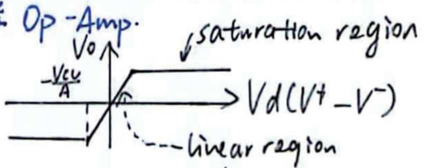




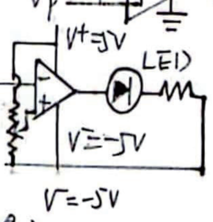
Ideal: A→∞

$R_i\to\infty$ $R_o\to0$

Open loop:

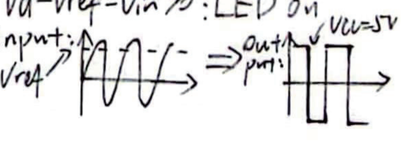


Application: comparer

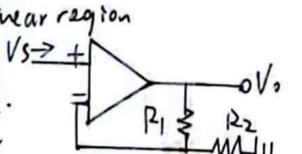


gain $g=\frac{V_o}{V_S}$

$V_d=V_{ref}-V_{in}>0$: LED on



接Vn
↓
负反馈：$k=\pm\frac{R_f}{R_s}$ 反馈电阻 / 输入电阻.



$$\frac{V_o}{V_S}=\frac{R_1+R_2}{R_2}$$

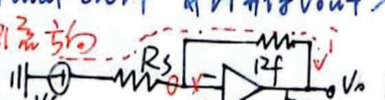
(非饱和前提)

Vo↑→Vn↑→Vp−Vn↓→Vo↓

Ideal: virtual open/virtual short ★计算得Vout>Vcc：假设错误(不满足linear region) Vout=±Vcc

★注意电势差+→−对应电流方向

反相(inverting)放大器：



$V_p=V_n$

$V_o=-\frac{R_f}{R_s}V_s$

{ 输出电流求和 该点不用KCL "虚短"失效 "虚断"

Inverse Laplace：部分分式展开. 分子为0：零点 分母为0：极点

/待定系数法

Case 1: 单极点 留数法. 去掉分母的某一项，根代入其它部分 eg. $\frac{1}{s(s+3)}=\frac{K_1}{s}+\frac{K_2}{s+3}$

$K_1=\frac{1}{0+3}=\frac{1}{3}$

$K_2=\frac{1}{-3}=-\frac{1}{3}$

Case 2: 重极点 求一阶导，找低一阶的系数/直接待定系数 留数法先求形式的项

Case 3: $\frac{A_1s+A_2}{s^2+As+B}=\frac{A_1(s+\alpha)+B_1\beta}{(s+\alpha)^2+\beta^2}\Rightarrow e^{-\alpha t}(A_1\cos\beta t+B_1\sin\beta t)$

$$\frac{K_1}{s+p}+\frac{K_2}{(s+p)^2}+\cdots+\frac{K_r}{(s+p)^r}\qquad K_{r-m}=\frac{1}{m!}\frac{d^m}{ds^m}\left((s+p)^rF(s)\right)\Big|_{s=-p}\qquad K_r=F(s)(s+p)^r\Big|_{s=-p}$$

★有运放电路接地点不能KCL(忽略了运放供电)

一阶线性方程: $y'+P(t)\cdot y=Q(t)$ 通解: $y=e^{-\int P(t)dt}\left(\int Q(t)\cdot e^{\int P(t)dt}dt+C\right)$

二阶线性齐次: $y''(t)+P\cdot y'(t)+Q\cdot y(t)=0$ 特征方程(characteristic equation): $\lambda^2+P\lambda+Q=0$

特征方程 $\Delta>0$: 通解为 $y=C_1e^{\lambda_1 t}+C_2e^{\lambda_2 t}$ $\Delta=0$: $y=(C_1+C_2\cdot t)e^{\lambda t}$ } C_1, C_2 由两个电路初始条件确定

$\Delta<0$: $\lambda_{1,2}=\alpha\pm\beta j$ $y=e^{\alpha t}(C_1\cos\beta t+C_2\sin\beta t)$

非齐次: 设特解. ① 非齐次项为 $P_m(t)e^{rt}$: $y^*=x^kQ_m(x)e^{rx}$ ← 多项式相同, 次数待定(无多项式因子设常数A ← e指数相同

e的指数r作为特征方程根的重数

② 非齐次项为 $e^{\alpha t}[P_l(t)\cos\beta t+Q_m(t)\sin\beta t]$ 令 $n=\max(l,m)$ $y^*=x^ke^{\alpha t}[M_n(t)\cos\beta t+N_n(t)\sin\beta t]$

(待定多项式系数 求导代回原始方程求解)

$(\alpha+\beta i)$是否为特征根

e指数相同 三角函数也系数相同.

指数/三角 × 多项式: 表格积分法.

$\int e^{at}\sin bt\,dt=\frac{1}{a^2+b^2}e^{at}(a\sin bt-b\cos bt)+C$

$\int e^{at}\cos bt\,dt=\frac{1}{a^2+b^2}e^{at}(b\sin bt+a\cos bt)+C$

$u \quad u' \quad u'' \quad \cdots \quad 0$ 多项式反复求导到0

$+ \quad - \quad +$

$v \quad \int v \quad \iint v \quad \cdots \quad \iint\cdots v$ 指数/三角反复积分

电容电压不突变 电感电流不突变

稳态: 电容开路 电感短路

$t=0^-\Rightarrow t=0^+$(初值) 任意t(微分方程)

$t\to\infty$: 一切归零(无外加源)

一阶电路全响应: $\boxed{\text{三要素法}}$

※注意电容i是V的导数, 电感V是i的导数.

注意Thevenin/Norton的应用(方便使用三要素法)

电容: $C=\frac{\varepsilon S}{4\pi kd}$ $q=C\cdot V$ $\boxed{i(t)=C\cdot\frac{dv(t)}{dt}\quad W=\frac{1}{2}CV^2}$

串并联规律与电阻相反(并联更大, 串联更小)

电感: $L=\frac{N^2\mu S}{K}$ $\lambda=Li$ $\boxed{v(t)=L\cdot\frac{di(t)}{dt}\quad W=\frac{1}{2}Li^2}$

串并联规律与电阻相同(串联更大, 并联更小)

RC电路: $V_C(t)=V_C(0^+)\cdot e^{-\frac{t}{\tau}}$ ($\tau=RC$ time constant 单位s)

RL电路: $i_L(t)=I_S(0^+)\cdot e^{-\frac{t}{\tau}}$ ($\tau=\frac{L}{R}$)

※R是Req(等效)

阶跃响应(RC为例): $V(t)=V_C(0^+)\cdot e^{-\frac{t}{\tau}}+V_S\cdot(1-e^{-\frac{t}{\tau}})$ }

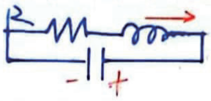
输入电压 $V_C(0^+)\to V_S$

$\to V_{末}-\Delta V\cdot e^{-\frac{t}{\tau}}$

$(V_{末}-V_{初})$

注意方向!

二阶串联RLC: (series)



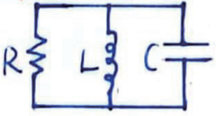
$$\frac{d^2V_C(t)}{dt^2}+\frac{R}{L}\cdot\frac{dV_C(t)}{dt}+\frac{1}{LC}V_C(t)=0$$

(Req)

$\alpha=\frac{R}{2L}$ $\omega_0=\frac{1}{\sqrt{LC}}$

※注意电容, 电感题目标定的电流/电压方向可能与实际情况/方程不一致

二阶并联RLC: (parallel)



(Req)

$$\frac{d^2i_L(t)}{dt^2}+\frac{1}{RC}\cdot\frac{di_L(t)}{dt}+\frac{1}{LC}\cdot i_L(t)=0$$

$\alpha=\frac{1}{2RC}$ $\omega_0=\frac{1}{\sqrt{LC}}$

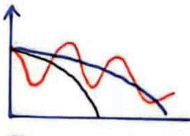
※先求特解 再求齐次通解中的待定常数(结合电路条件)

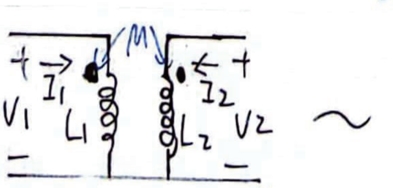
General Solution: $s=-\alpha\pm\sqrt{\alpha^2-\omega_0^2}$

Overdamped 过阻尼: $\alpha>\omega_0$ $y(t)=A_1e^{s_1t}+A_2e^{s_2t}$

Critically damped: $\alpha=\omega_0$ $y(t)=(A_1t+A_2)e^{-\alpha t}$

Underdamped: $\alpha<\omega_0$ $y(t)=e^{-\alpha t}[A_1\cos\omega_d t+A_2\sin\omega_d t]$, where $\omega_d=\sqrt{\omega_0^2-\alpha^2}$





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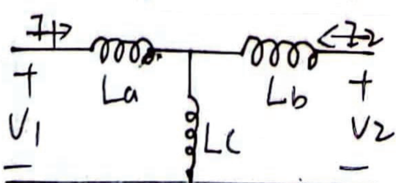
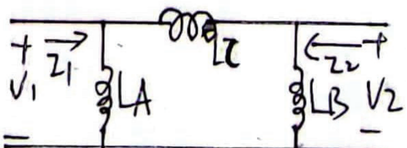


$L_a=L_1-M$

$L_b=L_2-M$

$L_C=M$

~



n^{th} derivative: $\mathcal{L}\left(\frac{d^n}{dt^n}f(t)\right)=s^nF(s)-\sum_{k=0}^{n-1}s^{n-1-k}f^{(k)}(0^-)$

Time periodicity:

if $f(t)=f(t+nT)$: $\mathcal{L}(f(t))=\frac{F_1(s)}{1-e^{-sT}}$ where $f_1(t)=\begin{cases}f(t) & 0<t<T\\ 0 & t>T\end{cases}$

Linearity: $\mathcal{L}(a_1f_1(t)+a_2f_2(t))=a_1\mathcal{L}(f_1(t))+a_2\mathcal{L}(f_2(t))$

$L_A=\frac{L_1L_2-M^2}{L_2-M}$ $L_B=\frac{L_1L_2-M^2}{L_1-M}$

$L_C=\frac{L_1L_2-M^2}{M}$

$\sin\alpha+\sin\beta=2\sin\frac{\alpha+\beta}{2}\cos\frac{\alpha-\beta}{2}$

$\sin\alpha-\sin\beta=2\cos\frac{\alpha+\beta}{2}\sin\frac{\alpha-\beta}{2}$

$\cos\alpha+\cos\beta=2\cos\frac{\alpha+\beta}{2}\cos\frac{\alpha-\beta}{2}$

$\cos\alpha-\cos\beta=-2\sin\frac{\alpha+\beta}{2}\sin\frac{\alpha-\beta}{2}$

RMS: $\sqrt{\frac{1}{T}\int_0^T i^2(t)\,dt}$

$\cos\varphi=\frac{e^{j\varphi}+e^{-j\varphi}}{2}$　$\sin\varphi=\frac{e^{j\varphi}-e^{-j\varphi}}{2j}$

$A\cos\omega t+B\sin\omega t=C\cos(\omega t-\theta)$　$C=\sqrt{A^2+B^2}$　$\theta=\arctan\frac{B}{A}$

$e^{\pm j\varphi}=\cos\varphi\pm j\sin\varphi$　中 $\omega=2\pi f$ 注意区分

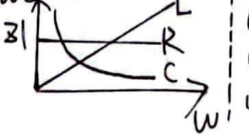
phasor domain 条件: 正弦,单频,线性定常 LTI (二极管 ×)

lag: 落后 lead: 领先　感性 ($\mathrm{Im}(Z)>0$): lag　容性 ($\mathrm{Im}(Z)<0$): lead

正弦波: $V_{rms}=\frac{\sqrt{2}}{2}V_m$　sin 半波: $V_{rms}=\frac{1}{2}V_m$ (与DC V/I 功率相等

phasor: $v(t)\to\hat{V}$　$\frac{dv}{dt}=j\omega\hat{V}$　$\int v\,dt=\frac{\hat{V}}{j\omega}$

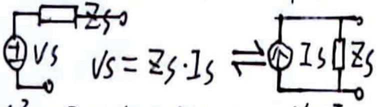
resistor $\hat{V}=R\cdot\hat{I}$　inductor $\hat{V}=j\omega L\cdot\hat{I}$　capacitor $\hat{V}=\frac{1}{j\omega C}\cdot\hat{I}$



阻抗 Z impedance
导纳 Y admittance
电阻 resistance R　$Z=R+jX$
电抗 reactance X
电导 conductance G　$Y=G+jB$
电纳 susceptance B

KCL与KVL ✓　注意: $\sin\angle 0=\cos\angle-90^\circ$ 需先化成cos　中 不同频率不能叠加

source transformation:



$V_s=Z_s\cdot I_s$　$I_s=\frac{V_s}{Z_s}$

Thevenin / Norton　V_{oc}/Z_{TH} 上一页

Instantaneous power 瞬时功率　$p=v(t)\cdot i(t)=\frac{V_mI_m}{2}\cos(\theta_v-\theta_i)+\frac{V_mI_m}{2}\cos(2\omega t+\theta_v+\theta_i)$

Complex power　$S=P+jQ=\frac{1}{2}V_m\cdot(I_m^*)=V_{rms}\cdot(I_{rms}^*)=|V_{rms}||I_{rms}|\angle(\theta_v-\theta_i)$　unit: VA

Real power = average power = P　unit: W　$P=\mathrm{Re}(S)=\frac{V_mI_m}{2}\cos(\theta_v-\theta_i)$

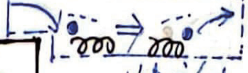
Reactive power 无功功率 Q　unit: VAR　$Q=\mathrm{Im}(S)=\frac{V_mI_m}{2}\sin(\theta_v-\theta_i)$

power factor　$PF=\frac{P}{S_a}=\cos(\theta_v-\theta_i)$

Apparent power 视在功率　$S_a=|S|=\sqrt{P^2+Q^2}$　unit: VA

火线 Line　零线 neutral　地线 ground

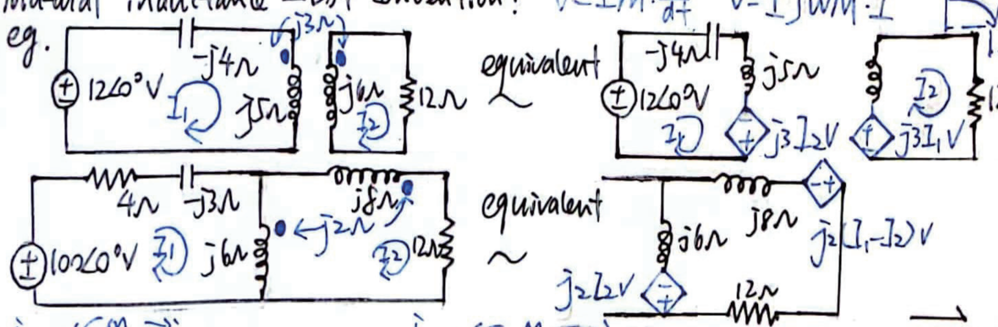
Mutual inductance — Dot convention:　$v=\pm M\cdot\frac{di}{dt}$　$\hat{V}=\pm j\omega M\cdot\hat{I}$



产生的互感电动势方向

注意方向!

eg.

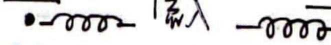


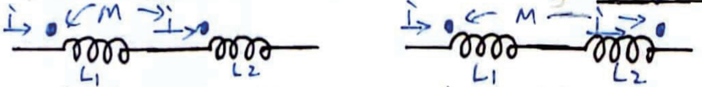
Energy:

$W=\frac{1}{2}L_1i_1^2+\frac{1}{2}L_2i_2^2\pm M\cdot i_1i_2$

+: 磁通量相互增强　—: 抵消

(i_1, i_2 都是流入点/流出点)





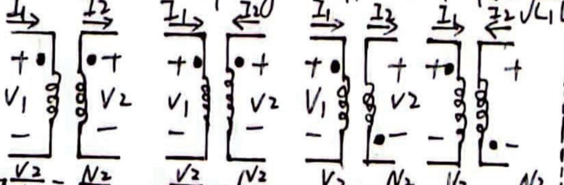
$L=L_1+L_2+2M$　　$L=L_1+L_2-2M$

Coupling coefficient　$K=\frac{M}{\sqrt{L_1L_2}}$ $(M\geq 0)$

$0\leq K\leq 1$

Ideal transformer 无漏磁 $\mu\to\infty$ $K=1$ $R_1=R_2=0$

$\frac{V_1}{V_2}=\frac{N_1}{N_2}$　$\frac{I_1}{I_2}=\frac{N_2}{N_1}$ (反比)　$Z_{eq}=Z_{load}\cdot\left(\frac{N_{primary}}{N_{secondary}}\right)^2$



$\frac{V_2}{V_1}=\frac{N_2}{N_1}$　$\frac{I_2}{I_1}=\frac{N_1}{N_2}$

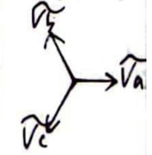
$\frac{V_2}{V_1}=\frac{N_2}{N_1}$　$\frac{I_2}{I_1}=-\frac{N_1}{N_2}$

$\frac{V_2}{V_1}=-\frac{N_2}{N_1}$　$\frac{I_2}{I_1}=-\frac{N_1}{N_2}$

$\frac{V_2}{V_1}=-\frac{N_2}{N_1}$　$\frac{I_2}{I_1}=\frac{N_1}{N_2}$

Balanced 3-phase system — Same magnitude / frequency 中 different phase (120°) Same impedance

$\hat{V}_a+\hat{V}_b+\hat{V}_c=$



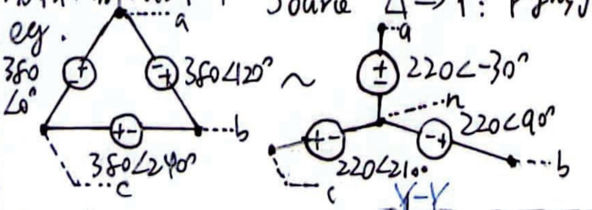
Y形连接的中性点 (V_n) 电压为0 (没有实际的中性点可以连起来)

分析: 都化为 Y-Y　Source Δ→Y: 除以$\sqrt{3}$ 相位提前30°

(或乘以 $\left(\frac{1}{2}-j\frac{\sqrt{3}}{6}\right)$)

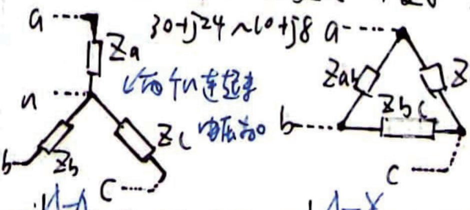
关于三相电路: 两相会触电 ≥3不会

eg.



Load Δ→Y: Δ更大 Y更小 $Z_Y=\frac{1}{3}Z_\Delta$

30+j24 ~ 10+j8



(没有n连起来 电压为0

Line voltage/current: 每条导线上的

phase voltage/current: 每两相之间的 电压(小写)

假设abc正相序　负载: 大写　Vp: 电源电压

a: ∠0°　b: ∠-120°　c: ∠-240°

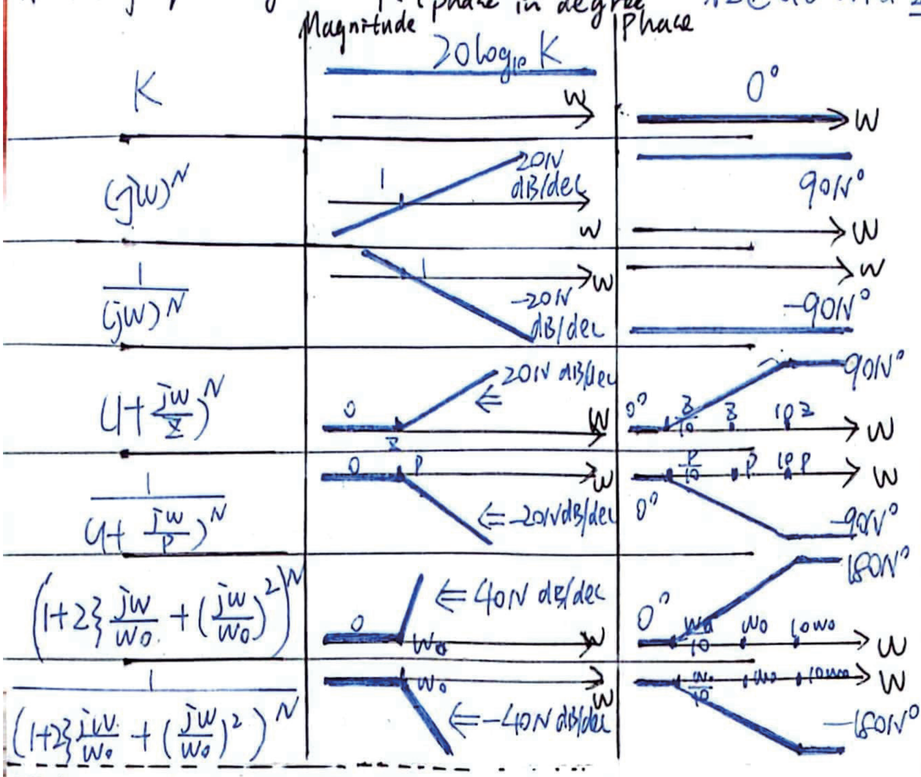
	Y-Y	Y-Δ	Δ-Δ	Δ-Y
相	$V_{an}=V_p\angle 0^\circ$ $V_{bn}=V_p\angle-120^\circ$ 同样电流 $V_{cn}=V_p\angle-240^\circ$	$V_{an}=V_p\angle 0^\circ$ $I_{AB}=V_{AB}/Z_\Delta$ $I_{BC}=V_{BC}/Z_\Delta$　$I_{CA}=\frac{V_{CA}}{Z_\Delta}$	$V_{ab}=V_p\angle 0^\circ$ $V_{bc}=V_p\angle-120^\circ$ $I_{AB}=V_{ab}/Z_\Delta$　$I_{BC}=\frac{V_{bc}}{Z_\Delta}$　$I_{CA}=\frac{V_{ca}}{Z_\Delta}$	$V_{ab}=V_p\angle 0^\circ$ $V_{bc}=V_p\angle-120^\circ$ 同样电流 $V_{ca}=V_p\angle-240^\circ$
线	$V_{ab}=\sqrt{3}V_p\angle 30^\circ$ $V_{bc}=V_{ab}\angle-120^\circ$ $V_{ca}=V_{ab}\angle-240^\circ$ $I_a=V_{an}/Z_Y$ $I_b=I_a\angle-120^\circ$　$I_c=I_a\angle-240^\circ$	$V_{ab}=V_{AB}=\sqrt{3}V_p\angle 30^\circ$ $V_{bc}=V_{BC}=V_{ab}\angle-120^\circ$ $I_a=I_{AB}\sqrt{3}\angle-30^\circ$ $I_b=I_a\angle-120^\circ$	$I_a=I_{AB}\sqrt{3}\angle-30^\circ$ $I_b=I_a\angle-120^\circ$ $I_c=I_a\angle-240^\circ$ 同相电压	$I_a=V_p\angle-30^\circ/(\sqrt{3}Z_Y)$ $I_b=I_a\angle-120^\circ$ $I_c=I_a\angle-240^\circ$

分子: numerator 分母: denominator 传递函数H(w)单位量纲(Ω/mho) 若输入有多个频率：每个分响应叠加

Bode Plots (Diagram): Bel: $G_B = \log_{10}\left(\frac{P_1}{P_2}\right)$ Decibel: $G_{dB} = 10\log_{10}\left(\frac{P_1}{P_2}\right) = 20\log_{10}\left(\frac{V}{V_{ref}}\right) = 20\log_{10}\left(\frac{I}{I_{ref}}\right)$

X-axis: freq. in logscale Y: $\begin{cases}\text{magnitudes in dB} \\ \text{phase in degree}\end{cases}$

传递函数拆成乘积 Bode Plot 直接相加 $2X \Rightarrow 2+X$


Magnitude
Phase
K
20log10 K
0°
w
$(jw)^N$
1
20N dB/dec
90N°
$\frac{1}{(jw)^N}$
-20N dB/dec
-90N°
$(1+\frac{jw}{z})^N$
0
z
20N dB/dec
0°
z/10
z
10z
90N°
$\frac{1}{(1+\frac{jw}{p})^N}$
0
p
-20N dB/dec
0°
p/10
p
10p
-90N°
$\left(1+2\xi\frac{jw}{w_0}+\left(\frac{jw}{w_0}\right)^2\right)^N$
0
w_0
40N dB/dec
0°
$w_0/10$
w_0
$10w_0$
180N°
$\frac{1}{\left(1+2\xi\frac{jw}{w_0}+\left(\frac{jw}{w_0}\right)^2\right)^N}$
-40N dB/dec
-180N°

$$eg.\ H(w) = \frac{200jw}{(jw+2)(jw+10)}$$

$$= 200\cdot(jw)\cdot\frac{1}{jw+2}\cdot\frac{1}{jw+10}$$

▷Resonance Def: $\mathrm{Im}(Z)=0$

(center frequency: H(w) 模值最大点)

I和Vs同相 $|V_L| = |V_C|$ 可远大于Vs

串联RLC：阻抗最小 电流最大 LC电压抵消

并联RLC：阻抗最大 电流最小 LC电流抵消

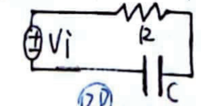
	in series	in parallel
谐振频率 w_0	$\frac{1}{\sqrt{LC}}$	$\frac{1}{\sqrt{LC}}$
带宽B	$\frac{w_0}{Q}$	$\frac{w_0}{Q}$
品质因数Q	$\frac{w_0L}{R} = \frac{1}{w_0RC}$	$\frac{R}{w_0L} = w_0RC$
半功率点 w_1, w_2	$w_0\sqrt{1+\left(\frac{1}{2Q}\right)^2} \pm \frac{w_0}{2Q}$	$w_0\sqrt{1+\left(\frac{1}{2Q}\right)^2} \pm \frac{w_0}{2Q}$
$Q \geq 10$	$w_0 \pm \frac{B}{2}$	$w_0 \pm \frac{B}{2}$

半功率点定义：$H(w) = \frac{\sqrt{2}}{2}H_{max}(w)$

Filter. Low-pass (LP) High-pass (HP) Band-pass (BP) Band-reject (BR)

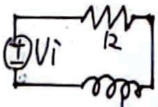
判断通带和阻带：半功率点

一阶RC: $H(w) = \frac{V_o}{V_i}$



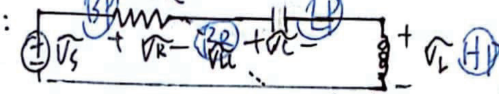
对R：高通 对C：低通

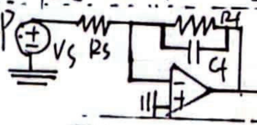
一阶RL:



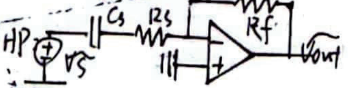
对R：低通 对L：高通

二阶RLC:


BP
LP
Vs
+ VR −
BR
VLC
+ VC −
+
VL
HP
−


LP
Vs
Rs
Rf
Cf
Vout

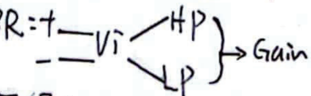
$$\frac{V_{out}}{V_S} = -\frac{R_f}{R_S}\cdot\frac{1}{1+jw/w_C} \quad w_C = \frac{1}{R_fC_f}$$


HP
Vs
Cs
Rs
Rf
Vout

$$\frac{V_{out}}{V_S} = -\frac{R_f}{R_S}\cdot\frac{jw/w_C}{1+jw/w_C} \quad w_C = \frac{1}{R_SC_S}$$

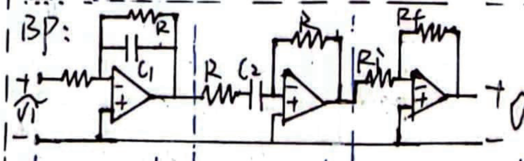
Cascading active filters (BP: in series; BR: in parallel)

串联：传递函数相乘 并联：相加


BR:
+
−
Vi
HP
LP
Gain

$$H(w) = -\frac{R_f}{R_I}\left(-\frac{1}{1+jw/w_1} - \frac{jw/w_2}{1+jw/w_2}\right)$$

必须 $w_{HP} > w_{LP}$ 否则无通带


BP:
R
C1
R
C2
R
Rf
Ri
+
Vi
−
+
Vo
−

LP $w_2 = \frac{1}{RC_1}$ | HP $w_1 = \frac{1}{RC_2}$ | Gain $K = -\frac{R_f}{R_i}$

$$\frac{V_o}{V_i} = \left(-\frac{1}{1+jw/w_2}\right)\left(-\frac{jw/w_1}{1+jw/w_1}\right)\left(-R_f/R_i\right)$$

Laplace transformation: $\int_{0^-}^{\infty} f(t)e^{-st}\,dt = F(s)$ $s = \sigma + jw$ 复频率

也不要忘记三要素法!

$1\cdot u(t) \to \frac{1}{s}$ $e^{-at}u(t) \to \frac{1}{s+a}$ $e^{-(a+bj)t}u(t) = \frac{1}{s+a+bj}$ $\cos(at)\cdot u(t) \to \frac{s}{s^2+a^2}$ $te^{t}u(t) \to \frac{1}{(s-1)^2}$

$\sin(at)u(t) \to \frac{a}{s^2+a^2}$ $u(t-T) \to \frac{e^{-sT}}{s}$ $t\cdot u(t) \to \frac{1}{s^2}$ $t^n\cdot u(t) \to \frac{n!}{s^{n+1}}$ $te^{-t}u(t) \to \frac{1}{(s+1)^2}$

$e^{at}\cos bt\cdot u(t) \to \frac{s-a}{(s-a)^2+b^2}$ $e^{at}\sin bt\cdot u(t) \to \frac{b}{(s-a)^2+b^2}$ $t\sin t\,u(t) \to \frac{2s}{(s^2+1)^2}$ $\delta(t) \to 1$

$t\cos t\,u(t) \to \frac{s^2-1}{(s^2+1)^2}$ $\frac{\sin t}{t}u(t) \to \arctan\left(\frac{1}{s}\right)$ $\frac{\cos(t)}{t}u(t) \to \frac{1}{2}\ln\left(1+\frac{1}{s^2}\right)$

$\sinh(t)\cdot u(t) \to \frac{1}{s^2-1}$ $\cosh(t)\cdot u(t) \to \frac{s}{s^2-1}$ $\frac{t^n}{n}u(t) \to \frac{(n-1)!}{s^{n+1}}$ $\delta(t-t_0) = e^{-t_0s}$

$\mathcal{L}(t^ne^{at}\sin bt) = \frac{n!\cdot b}{((s-a)^2+b^2)^{n+1}}$ $\mathcal{L}(t^ne^{at}\cos bt) = \frac{n!(s-a)}{((s-a)^2+b^2)^{n+1}}$ $(n \geq 0)$

$\mathcal{L}\left(\frac{d}{dt}f(t)\right) = sF(s) - f(0^-)$ $\mathcal{L}\left(\int_0^t f(\tau)d\tau\right) = \frac{F(s)}{s}$ $\mathcal{L}(f(t-\tau)) = e^{-s\tau}F(s)\ (\tau > 0)$

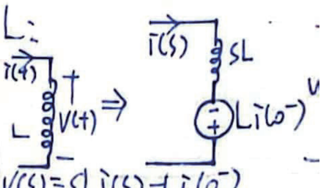
$\mathcal{L}(e^{-at}f(t)) = F(s+a)$ $\mathcal{L}(tf(t)) = -\frac{d}{ds}F(s)$ $\mathcal{L}\left(\frac{f(t)}{t}\right) = \int_0^{\infty}F(\tau)d\tau$

$\mathcal{L}(f(at)) = \frac{1}{a}\cdot F\left(\frac{s}{a}\right)$ Initial value: $f(0^-) = \lim_{s\to\infty} sF(s)$ Final value: $f(\infty) = \lim_{s\to 0} sF(s)$

equivalent circuit:

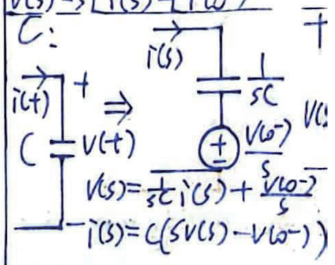
R: $V(s) = R\cdot I(s)$

L:


i(t)
+
L
V(t)
−
i(s)
sL
L i(0⁻)

$V(s) = sL\,i(s) - L\,i(0^-)$

C:


i(t)
+
C
V(t)
−
i(s)
$\frac{1}{sC}$
$\frac{V(0^-)}{s}$

$V(s) = \frac{1}{sC}i(s) + \frac{V(0^-)}{s}$

$i(s) = C(sV(s) - V(0^-))$